



Republic of Rwanda
Ministry of Education



RTB | RWANDA
TVET BOARD

LEVELING SURVEYS APPLICATION

LSVLS302

Perform leveling surveys

Competence

RQF Level: 3

Learning Hours



100

Credits: 10

Sector: Construction and Building Services

Trade: Land surveying

Module Type: Specific

Curriculum: CBSLSV3001- TVET Certificate 3 in Land surveying

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Purpose statement	This module titled leveling surveys application was developed for students of RQF Level 3 land surveying for the purpose of equipping them with the skills, knowledge, and attitudes required to apply leveling techniques in projects related to measuring elevations. At the end of this module, the learner will be able to describe methods of leveling, perform instrument adjustments, and apply leveling techniques to determine elevations or differences in elevations of points and/or to establish points at various given elevations.					
Leaning assumed to be in place	Safety, Health and environment at workplace; Surveying tools and equipment; Surveying communication language; Linear distance measurement; Construction drawings interpretation; Building setting out; Angular measurements.					
Delivery modality	Training delivery		100%	Assessment		Total 100%
	Theoretical content		30%	Formative assessment	30%	50%
	Practical work:		70%		70%	
	• Group project and presentation	20%				
	• Individual project /Work	50%				
			Summative Assessment		50%	

Elements of Competency and Performance Criteria

Elements of competency	Performance criteria
1. Describe methods of levelling	1.1. Levelling terms are properly described.
	1.2. Direct Levelling methods are properly described according to their use.
	1.3. Indirect Levelling methods are properly described according to their use.
2. Perform levelling	2.1. Levelling instruments are properly identified according to

instrument adjustments	their types.
	2.2. Parts of levelling instruments are well described according to the types of instruments.
	2.3. The levels are temporally adjusted according to their types.
	2.4. The levels are permanently adjusted according to their axes.
3. Apply levelling methods to real case projects	3.1. Levels are properly taken and booked according to the types of readings.
	3.2. Reduced levels (RLs) are correctly calculated according to the methods of booking.
	3.3. Profile levelling is properly applied according to their types.
	3.4. Levelling is properly applied according to the earthwork calculations.
	3.5. Levelling is properly applied according to the contour map generation.
	3.6. Levelling errors are correctly adjusted according to their Sources.

Learning outcomes	At the end of the module the learner will be able to: 1. Describe methods of levelling 2. Perform levelling instrument adjustments 3. Apply levelling methods to real case projects
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Learning outcome 1: Describe methods of leveling	Learning hours: 15
Indicative content	
* Description of levelling terms (Refer to ISO 4463-2) • Levelling, level surface/ level line, horizontal plane/ horizontal line, vertical line, vertical datum, mean Sea Level, Objective of levelling, change-point, benchmark, back sight, foresight, intermediate sight. * Description of direct levelling methods • Simple levelling • Differential levelling	

• Fly levelling • Check levelling • Profile levelling • Reciprocal levelling • Cross levelling * Description of indirect levelling methods • Trigonometric levelling • Barometric levelling • Hypsometric levelling	
Resources required for the learning outcome	
Equipment	Levels, PPE, whiteboard.
Materials	Pen, notebooks, pieces of chalk, marker pens, pegs, station mark etc.
Tools	Staff reading, tripod, plumb bob, hammer, Tape measure, etc.
Facilitation techniques	• Demonstration and simulation • Individual and group work • Practical exercise • Trainer guided • Group discussion
Formative assessment methods	• Written assessment • Oral assessment • Performance assessment • Product based assessment

Learning outcome 2: Perform levelling instrument adjustments	Learning hours: 20
Indicative content	
* Types of levelling instrument (Refer to ISO 17123-2, ISO 1346, ISO 1141) • Dumpy level, • Electronic digital level, • Tilting level, • Automatic level	

* Parts of levelling instruments • Telescope, level tube, eyepiece, objective lens, objective lens focus (focusing knob), horizontal motion screw, circular level bubble, foot screws, target • Accessories of levelling instrument (Tripod, reading staff, plumb bob) * Temporary adjustment of level instrument • Setting up the instrument, • Centering, • Levelling up the instrument, • Focusing * 3. Permanent adjustment of the level • Two pegs Test • Adjustment of Horizontal & vertical axes and line of collimation	
Resources required for the indicative content	
Equipment	Level, PPE, whiteboard
Materials	Pen, notebooks, books, pieces of chalk, marker pens, pegs
Tools	Staff reading, tripod, plumb bob, hammer, tape measure, pegs
Facilitation techniques	• Demonstration and simulation • Individual and group work • Practical exercise • Individualized • Trainer guided • Group discussion
Formative assessment methods	• Written assessment • Oral assessment • Performance assessment • Product based assessment

Learning outcome 3: Apply levelling methods to real case projects	Learning hours: 65
Indicative content	

* **Types of readings**

- Back Sight (BS),
- Intermediate Sight (IS),
- Fore Sight (FS)}

* **Methods of booking**

- Rise and fall method,
- Plan of collimation/height of instrument method

* **Types of profiles**

- Longitudinal profile (elements, drawing),
- Cross-section (elements, drawing)

* **Collection of levelling data for Earthwork calculations**

- Selection of instruments and tools
- Collection of data (process)
- Data processing

* **Collection of levelling data for contour map generation (Refer to BS ISO 17123-5)**

- Grid levelling
- Interpolation of contours,
- Draw contour lines,

* **Levelling errors**

- Sources of levelling errors
 - ✓ Instrumental errors (collimation error, parallax error, Staff graduation errors, loose tripod shoes,)
 - ✓ Observational or personal errors (staff holding, Errors in reading the staff, booking errors)
 - ✓ Natural errors (heat shimmer, wind, curvature and refraction,)
- Adjustment of levelling errors
 - ✓ Minimizing errors,
 - ✓ Adjusting collimation error

Resources required for the indicative content	
Equipment	Level, whiteboard, PPE
Materials	Pen, notebooks, pieces of chalk, marker pens, Pegs
Tools	Staff reading, tripod, plumb bob, hammer, tape measure.
Facilitation techniques	• Demonstration and simulation • Individual and group work • Practical exercise • Individualized • Trainer guided • Group discussion
Formative assessment methods	• Written assessment • Oral assessment • Performance assessment • Product based assessment

Integrated/Summative assessment

H2N Construction Ltd is a company that specialized in surveying, design and construction of roads. This company won a tender for the design and construction of a 50m access road from Kacyiru-Remera road to Bob's house. H2N Construction Ltd has to carry out profile levelling, reduce the levels by the rise and fall method, and produce the longitudinal profile of the road. However, this target is not yet achieved due to the lack of skilled personnel to conduct the longitudinal profile survey.

As skilled personnel, H2N Construction Ltd wishes to hire you to handle such a problem. All necessary equipment, tools and materials are available at the company office. The associated pieces of information that will help you to accomplish the task are the following:

- Mark every 5 metres with a peg or other convenient material

- The reduced levels of the starting point (0+00) and ending points (0+50) are given
- H2N Construction Ltd needs a report of survey results
- The task will be done within 6 hours: 4 hours for the profile survey and 2 hours for report elaboration.

Note: The practical assessment must be done in groups of two (2) students.

Resources

Tools	Staff reading, tripod, plumb bob, hammer, Tape measure, Tripod, Pegs
Equipment	Level, whiteboard, PPE
Materials/ Consumables	Pen, notebooks, books, pieces chalk, marker pens, pegs

Assessable outcomes	Assessment criteria (Based on performance criteria)	Indicator	Observation		Marks allocation
			Yes	No	
Learning outcome 2: Perform leveling instrument adjustments (40%)	2.1. Levelling instruments, tools and materials are well selected.	Instruments are selected.			4
		Tools are selected.			3
		Materials are selected.			3
	2.3. Level instrument is temporally adjusted.	Level is set up.			8
		Level is centered.			7
		Level is levelled up.			8
		Focusing is done.			7
	Learning outcome 3: Apply levelling methods to real	1.1. Levels (readings) are properly taken and booked	Backsight is booked.		
Intersight is booked.					5
Foresight is					5

case projects (60%)		booked.			
		Interval distance is respected.			5
	3.2. Reduced levels (RLs) are correctly Calculated.	Reduce levels are calculated.			15
		An arithmetic check is done.			5
	1.2. Profiles are properly produced	The longitudinal profile is produced.			10
		Survey results are well reported.			5
		Time is respected.			5
	Total marks		100		
Percentage Weightage		100%			
Minimum Passing line % (Aggregate): 70%					

References

1. Bhavikatti, S. (2010). Basic civil engineering: New Age International (P) Limited.
2. Chandra, A. (2005). Surveying problem solving with theory and objective type questions: New Age International (P) Ltd.
3. CHARLES, D and GHILANI, P. (2012). Elementary surveying 13th edition. Pennsylvania State.
4. Davis, Raymond E. and al. (1981). Surveying Theory and Practice, 6th ed. New York: McGraw-Hill
5. Schofield, W. (2001) W. Engineering Surveying. Theory and examination problems for students. Fifth Edition.
6. ISO 17123-2 Optics and optical instruments — Field procedures for testing geodetic and surveying instruments — Part 2: Levels
7. ISO 4463-3 Measurement Methods For Building — Setting-Out And Measurement — Part 3: Check-Lists For The Procurement Of Surveys And Measurement Services
8. ISO 17123-5 Optics And Optical Instruments — Field Procedures For Testing Geodetic and Surveying Instruments — Part 5: Total Stations